

Introduction to Machine Learning

Course General Information

- **Course Name:** Machine Learning Course Code: CS 450
- **Teaching Load:** 3 Credit Hours
- **Contact Hours:** 4 Hrs. (2 Lectures + 2 Practical)
- **Teaching and Learning Strategies:**
 - Lectures
 - Open Discussion
 - Presentations
 - Laboratory
 - Projects
- **Assessment Methods:**
 - Quizzes (weeks: 5 and 12)
 - Midterm Exam (Week 8)
 - Assignments and different activities participation
 - Lab reports and Practical exam
 - Final Written Exam



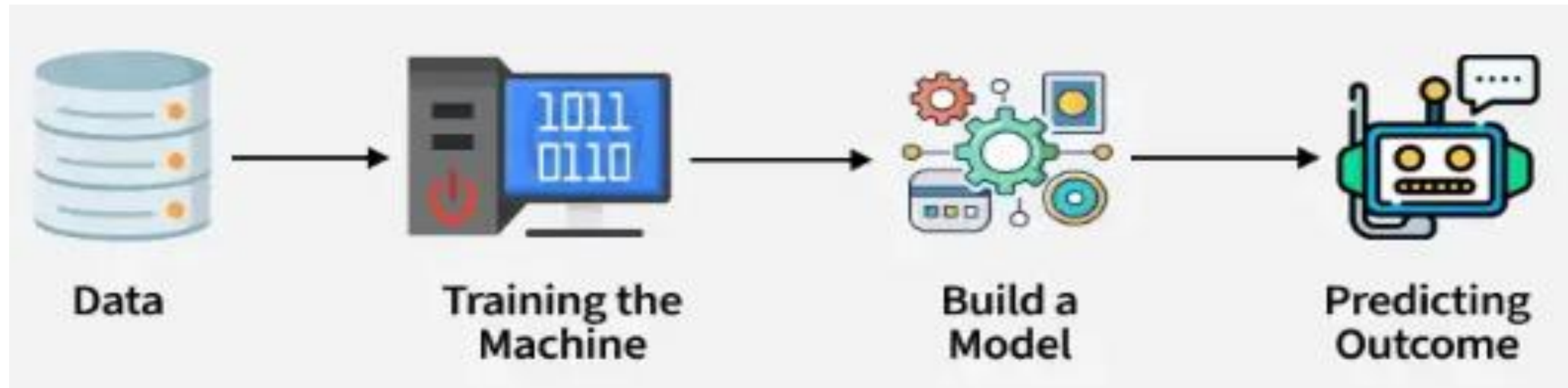
Course Plan

Auxiliary Materials

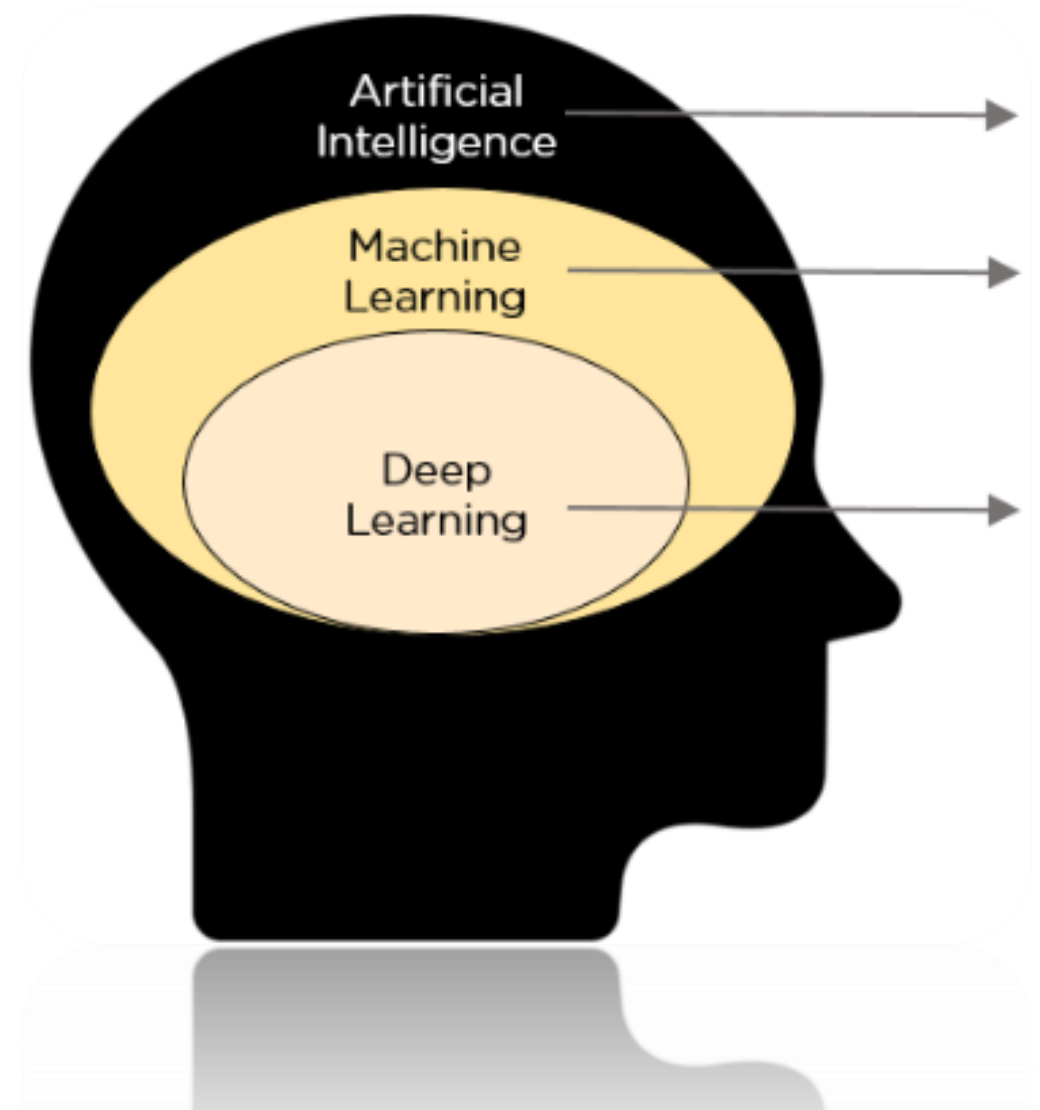
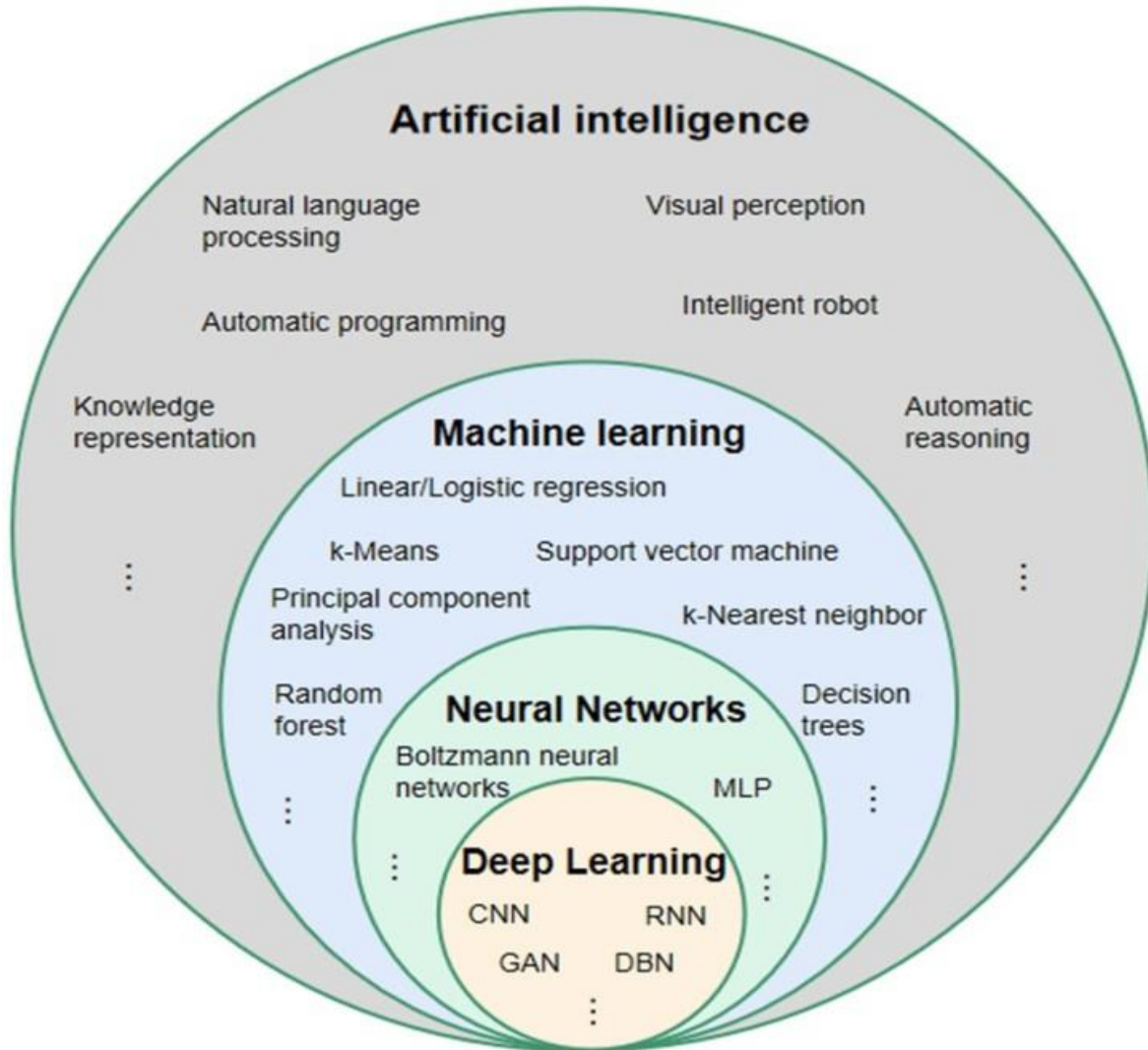
ML Videos

What is Machine Learning

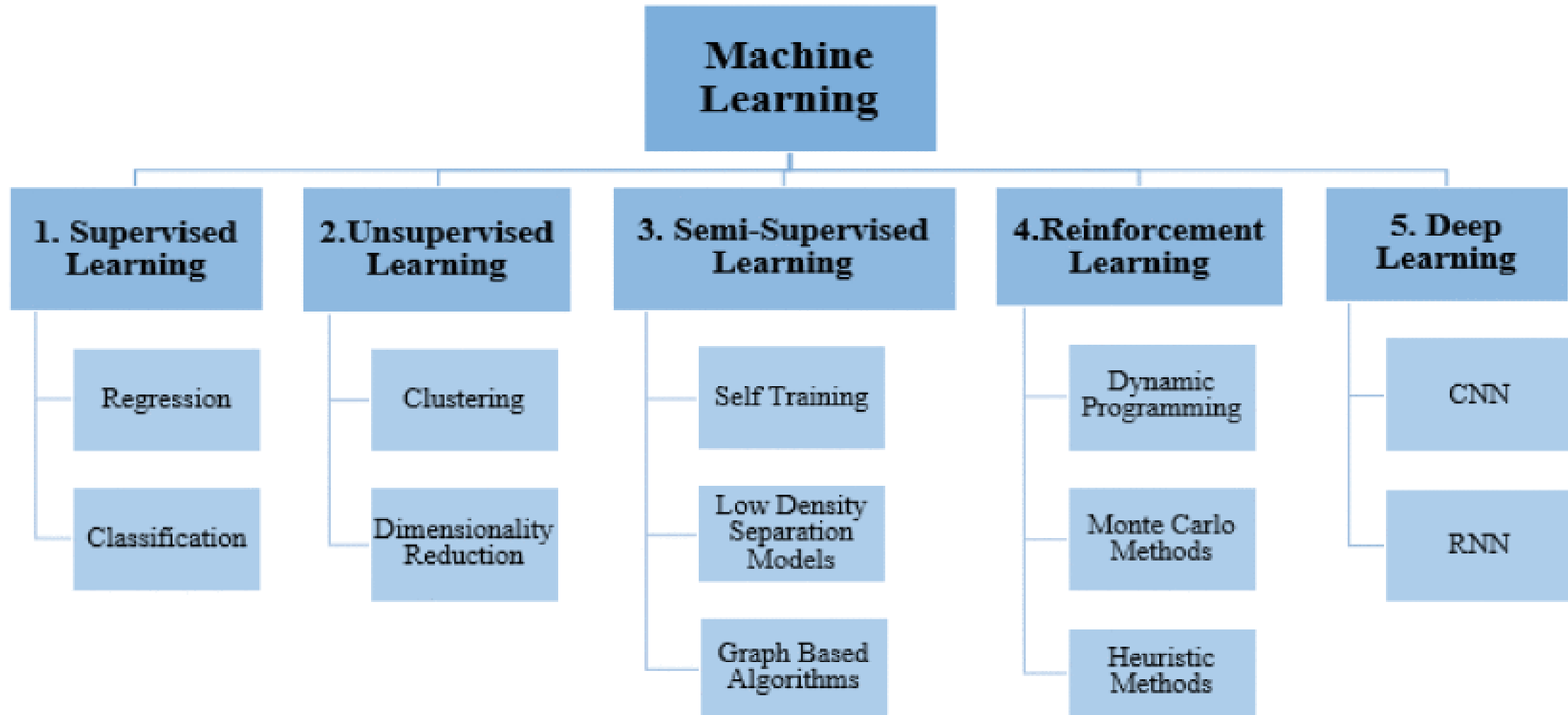
Machine learning (ML) allows computers to learn and make decisions without being explicitly programmed. It involves feeding data into algorithms to identify patterns and make predictions on new data. Machine learning is used in various applications, including image and speech recognition, natural language processing, and recommender systems.



Relationship between Machine LEARNING AND AI



Machine Learning Types and Algorithms.



Supervised Learning.

In general, supervised learning involves training datasets where data is presented **in pairs**: **an input** and **its corresponding correct output**.

An example of this is the actual price of electronic devices, which is directly affected by the features associated with each device. This process helps to train the algorithm and thus develop a model capable of predicting the price of new input devices that were not included in the training dataset



Unsupervised Learning



Semi-supervised Learning



Reinforcement learning

Reinforcement learning is different from supervised learning based on the way each method gains knowledge.

Supervised learning method learns from examples provided by an external supervisor.

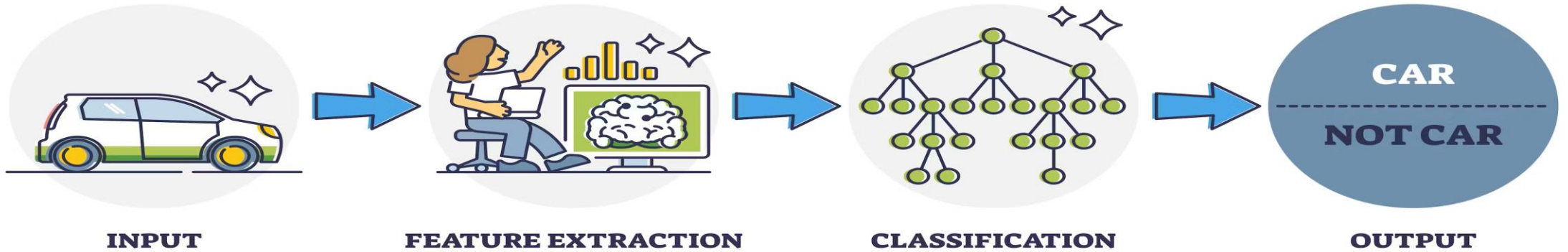
Whereas reinforcement learning uses direct interactions with the problem environment to gain knowledge

Examples:

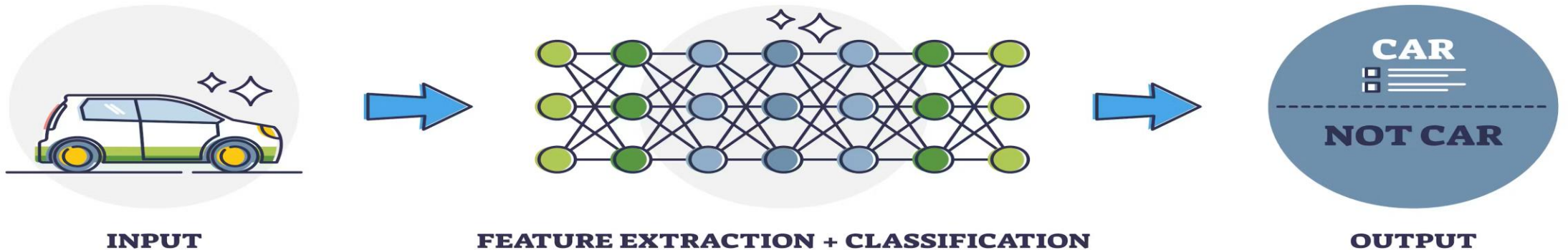
Dynamic Programming Monte Carlo methods Heuristic methods

Deep Learning

MACHINE LEARNING



DEEP LEARNING



Applications of ML

1. Solving Complex Business Problems such as:
 - Image and speech recognition in healthcare.
 - Language translation and emotion (sentiment) analysis.
2. Handling Large Volumes of Data:
 - Fraud (Fake) detection in financial transactions.
 - Social media platforms like Facebook and Instagram are predicting personalized feed recommendations from billions of interactions.
3. Automate Repetitive Tasks:
 - Email Filtering: Gmail uses ML to keep your inbox spam-free.
 - Chatbots: ML-powered chatbots resolve common issues like order tracking and password resets.
 - Data Processing: Automating large-scale invoice analysis for key insights.

Applications of ML (Cont.)

4. Personalized User Experience (Real-World Applications):

- Netflix: Suggests movies and TV shows based on viewing history.
- E-Commerce: Recommends products you're likely to purchase.

5. Self-Improvement in Performance

- Voice Assistants (e.g., Siri, Alexa): Learn user preferences, improve voice recognition, and handle diverse accents.
- Search Engines: Refine ranking algorithms based on user interactions.
- Self-Driving Cars: Enhance decision-making using millions of miles of data from simulations and real-world driving.

What Makes a Machine “Learn”?

A machine “learns” by recognizing patterns and improving its performance on a task based on data, without being explicitly programmed.

The Learning process involves the following

Data Input: Machines require data (e.g., text, images, numbers) to analyse.

Algorithms: Algorithms process the data, finding patterns or relationships.

Model Training: Machines learn by adjusting their parameters based on the input data using mathematical models.

Feedback Loop: The machine compares predictions to actual outcomes and corrects errors (via optimization methods like gradient descent).

Experience and Iteration: Repeating this process with more data improves the machine's accuracy over time.

Evaluation and Generalization: The model is tested on unseen data to ensure it performs well on real-world tasks.

Machine Learning Toolbox

The toolbox of machine learning consists of:

- 1- Data
- 2- Infrastructure
- 3- Algorithms

These tools differ in the case of:

- (I) Simple ML to
- (II) Advanced ML

I- Tools for Simple ML

1- Data is classified into



a) Structured:
Defined
Organized
tabled in a table

b) Unstructured:
Examples:
Images
videos
email message

Data Visualization:

Data Visualization may be required. Programs for data visualization, such as “Tableau” or Python libraries such as “Seaborn”

I- Tools for Simple ML (Cont.)

2- Infrastructure

- Computer (H/W)
- Web application (Jupyter Notebook)
- Programming Language (Python)
- ML Libraries (NumPy, Pandas, Scikit-Learn)
- Other languages may be used (C, C++) for Professional and advanced ML because they can use the GPU of the computer.

3- Algorithms:

- ❑ **For supervised Learning:** Linear regression – Logistic regression – Decision trees – KNN
- ❑ **For Unsupervised Learning:** K-Means Clustering

II- Tools for Advanced ML

1- Data:

If the data set size reaches Petabytes, it is described as a big data

2- Infrastructure for Advanced ML:

a) With big data, robust infrastructure is needed, such as:

- Distributed computing
- Cloud providers such as Amazon Web Services (AWS) or Google Cloud Platform.

These tools are used to process the data on virtual GPU

II- Tools for Advanced ML (Cont.)

2- Infrastructure for Advanced ML: (Cont.)

b) Programming Languages:

- ☐ C, C++.
- ☐ Python can also be used and compiled into C using the ML library TensorFlow from Google.
- ☐ Amazon Web Services, Microsoft Azure, Alibaba Cloud, and Google Cloud Platform offer pay-as-you-go GPU resources, starting with a free trial.

Note: Google announced in 2016 that it would publicly release a Tensor Processing Unit designed specifically for running TensorFlow.

II- Tools for Advanced ML (Cont.)

3- Algorithms for Advanced ML:

- Markov Models
- Support Vector Machines (SVM)
- Q- Learning (a type of Reinforcement Learning)
- Combinations of algorithms to create Unified mode (Ensemble modelling)
- Artificial Neural Networks (ANN)
- TensorFlow is used for Deep Learning (DL)
- Alternative libraries for NN include: Torch, Caffe, Keras

II- Tools for Advanced ML (Cont.)

3- Algorithms for Advanced ML (Cont.):

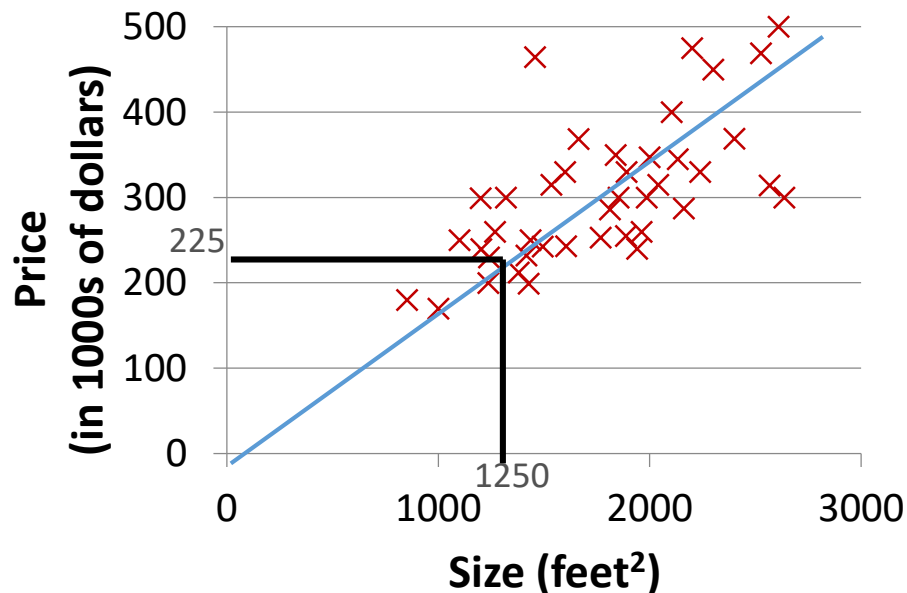
NOTES:

- Keras is an open-source DL library that sits on top of TensorFlow and other frameworks.
- Caffe is an open – source DL for image classification written in C++, but has a Python interface
- Torch is an open–source DL algorithm and functions using the Lua programming Language.

Regression Vs Classification

Regression

- Predict continuous valued output. (price of house)
- The output is an arbitrary value according to hypothesis.



Classification

- Predict discrete valued output. Breast cancer (malignant, benign)
- The output is 0 (negative class) or 1 (positive class).

